SENSOR-SUPPORTED SIMULATION-BASED APPROACH FOR HEALTHY, RESILIENT, AND SUSTAINABLE INDOOR ENVIRONMENTS

CONTRIBUTORS:

MR. NOOR-E SADMAN, ADJUNCT LECTURER, CSE, IUB

MS. NOUREEN ISLAM, ADJUNCT LECTURER, CSE, IUB

MR. TAHFIZUL HASAN ZIHAN, UNDERGRADUATE STUDENT, CSE, IUB

DR. MAHADY HASAN, ASSOCIATE PROFESSOR, CSE, IUB

DR. M A I KHAN, MIMA, LECTURER, ENVIRONMENTAL FLUID MECHANICS, UNIVERSITY OF LEEDS

DR. FERDOWS ZAHID, ASSOCIATE PROFESSOR, CSE, IUB

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BACKGROUND AND INTRODUCTION

This project is a collaboration between the Department of Computer Science and Engineering, Independent University Bangladesh (IUB), School of Computing, Electronics and Mathematics, University of Coventry, UK and School of Civil Engineering at the University of Leeds, UK. This project brings together expertise in environmental sensors, wireless networking, and fluid dynamics to develop a novel dynamic data-driven computational modelling approach to design and operate the next generation of healthy, smart and sustainable indoor environments. We will build on new approaches and capabilities developed at University of Leeds to now develop advanced real-time sensing approaches to adapt building ventilation for combined health, comfort, and energy objectives.

Indoor air quality (IAQ) is a critical factor in ensuring the health and well-being of individuals, especially in educational settings where occupants spend a significant portion of their day. With the rising concern over airborne infections, the need to optimize ventilation strategies becomes paramount. This report details an initial study conducted in a university classroom with air cooling in a hot and humid environment, representative of typical educational institutions in Bangladesh.

Numerous studies have addressed the importance of IAQ and its impact on occupants' health, with a particular focus on educational environments. However, the intersection of IAQ, thermal comfort, and infection risk mitigation through ventilation strategies in hot and humid climates remains a relatively unexplored area. Our research builds upon existing work by incorporating a holistic approach, combining sensor data collection and computational fluid dynamics (CFD) simulations.

The primary goal of our research is to understand the interplay between ventilation strategies, IAQ, thermal comfort, and infection risk in a challenging climatic context. By leveraging an array of sensors and advanced CFD modelling, we aim to assess the current state of the indoor environment and identify optimal ventilation strategies that balance energy efficiency, IAQ improvement, and infection risk mitigation. Our long-term goal will be to extend our current approach and apply it to the outdoor environment (specific localities in and around the cities in Bangladesh).

The goals and approaches of our study can be succinctly expressed by 5 action words: **Measure, Monitor, Assess, Design, Predict.**

Measure and monitor: We deployed an array of IoT sensor devices to measure and monitor the following indoor air quality parameters: temperature, humidity, air pressure and the concentrations of CO_2 and particulate matters (PM1.0, PM2.5 and PM10). These parameters are measured and monitored in real-time under real-life situations.

Assess: Analysis of the sensor data coupled with CFD simulations provide a detailed understanding of the dynamic indoor environment with changing occupancy and ventilation.

Design: A 3D CFD model was employed using the Reynolds averaged Navier-Stokes (RANS) approach, specifically the k-ω SST RANS model. This allowed us to simulate turbulent airflow, thermal conditions, and pollutant dispersion within the classroom. CFD simulations will enable us to understand the relationship between the classroom design, IAQ, thermal comfort and infection risk.

Predict: Combining sensor data and CFD simulation in conjunction with machine learning models will allow us to create data assimilation (DA)-based reliable reduced order models (ROMs) to predict the behavior of the dynamic indoor environment in the classroom. Thereby, providing us with a digital twin which could be used to predict and control the classroom indoor environment for improved IAQ and energy efficiency.

This report has been structured as follows: we provide a detailed description of the assembled IoT sensor devices along with the calibration results; then we present an analysis of the sensor data collected from a University classroom over a 8 month period; we then describe the CFD model and parameters employed in our study; our CFD simulation results have been presented in the next section; then we identify a few specific short term and long term goals for this project; a progress timeline has also been provided; finally we conclude by summarizing our achievements.

IoT SENSOR DEVICES

To build our own IoT sensor device we used 3 stand-alone sensor devices, a microcontroller and a development board. The 3 sensor components are: PMS5003 for measuring particulate matters (PM1.0. PM2.5 and PM10); BME280 sensor records temperature, humidity and air pressure and MH-19B is the CO_2 detector. The Arduino Mega 2560 WiFi is the microcontroller of the system and a development board with a facility of inbuilt wifi system, which helps users to transmit data easily and store it to a database for future research purposes. All the sensors are connected to the development board for transmission of data. The PMS5003 air quality sensor is connected with the Arduino mega development board through UART communication, its TX and RX are connected with the Arduino mega's RX and TX pins. The PMS5003 transmits data of PM2.5 to the Arduino mega. BME280 provides temperature, humidity and air pressure data to the Arduino mega using I2C communication. Its SDA and SCL pins are connected to Arduino mega's SDA and SCL pins. MH-Z19B is a CO2 detecting sensor, which operates in UART communication, and its RX and TX pins are connected with the Arduino's TX and RX pins.

Arduino Mega 2560 WiFi takes readings from the sensors. Arduino Mega connected with a PC sends all the sensor data to a MQTT Broker. A Google Sheet APIs runs on the Google Cloud, which stores the data in the cloud and google sheet file. The script then adds a timestamp with every single data and posts every single data simultaneously to the google cloud using RestAPIs and GoogleAPIs with very low latency. All the data is dynamically added to the google sheet automatically. Using the google sheet charts all the sensor data are visualized in different linear time series data charts according to the different obtained air quality parameters. The data from our sensor devices has been calibrated against an Industry standard top quality sensor device called AirVisual Pro. The calibration data is presented in Figure 2 for all 6 IAQ parameters – temperature, humidity, CO2, PM1.0, PM2.5 and PM10.

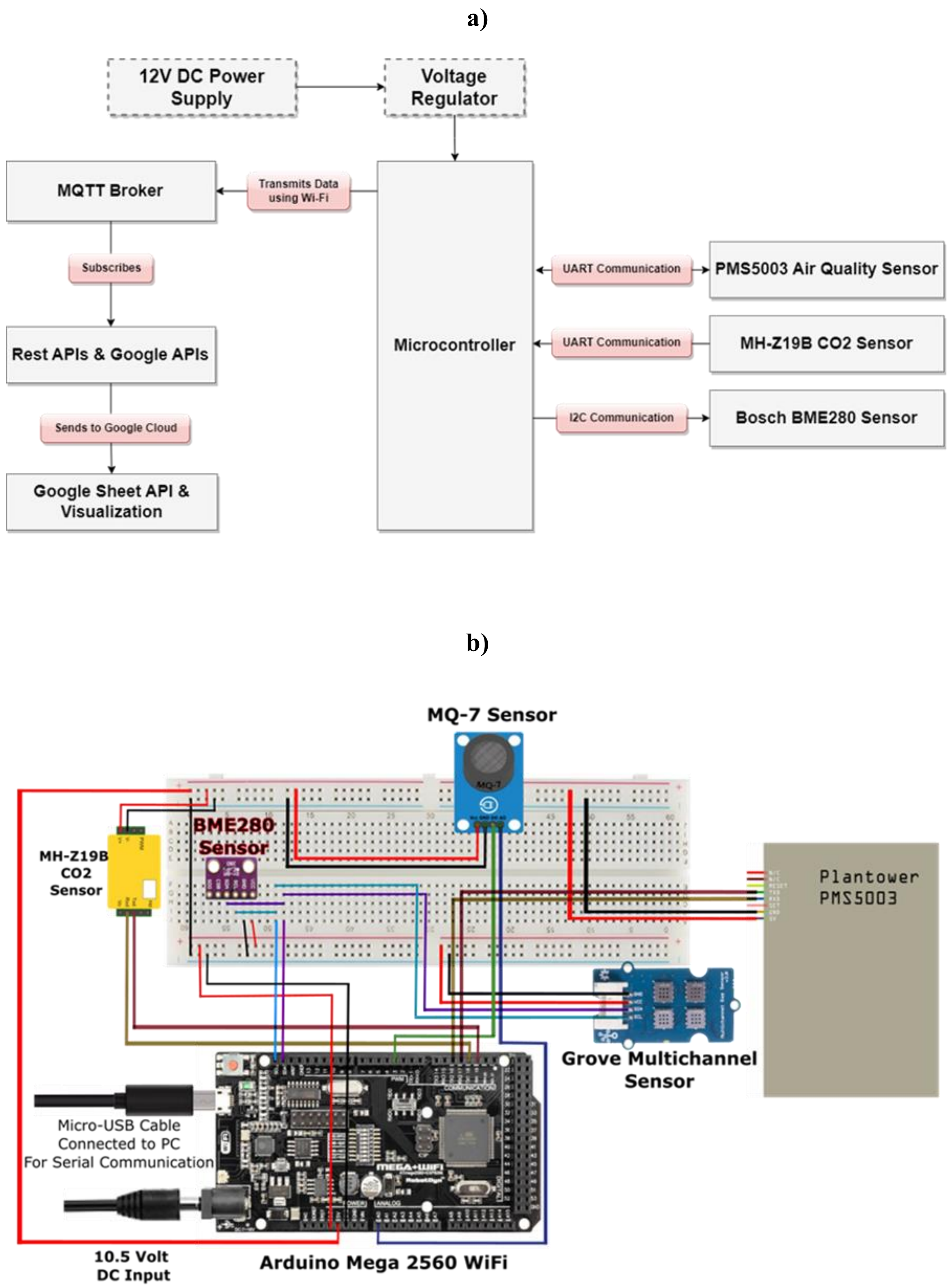


Figure 1: a) Block diagram and b) Schematic diagram of the IoT sensor device used in this study

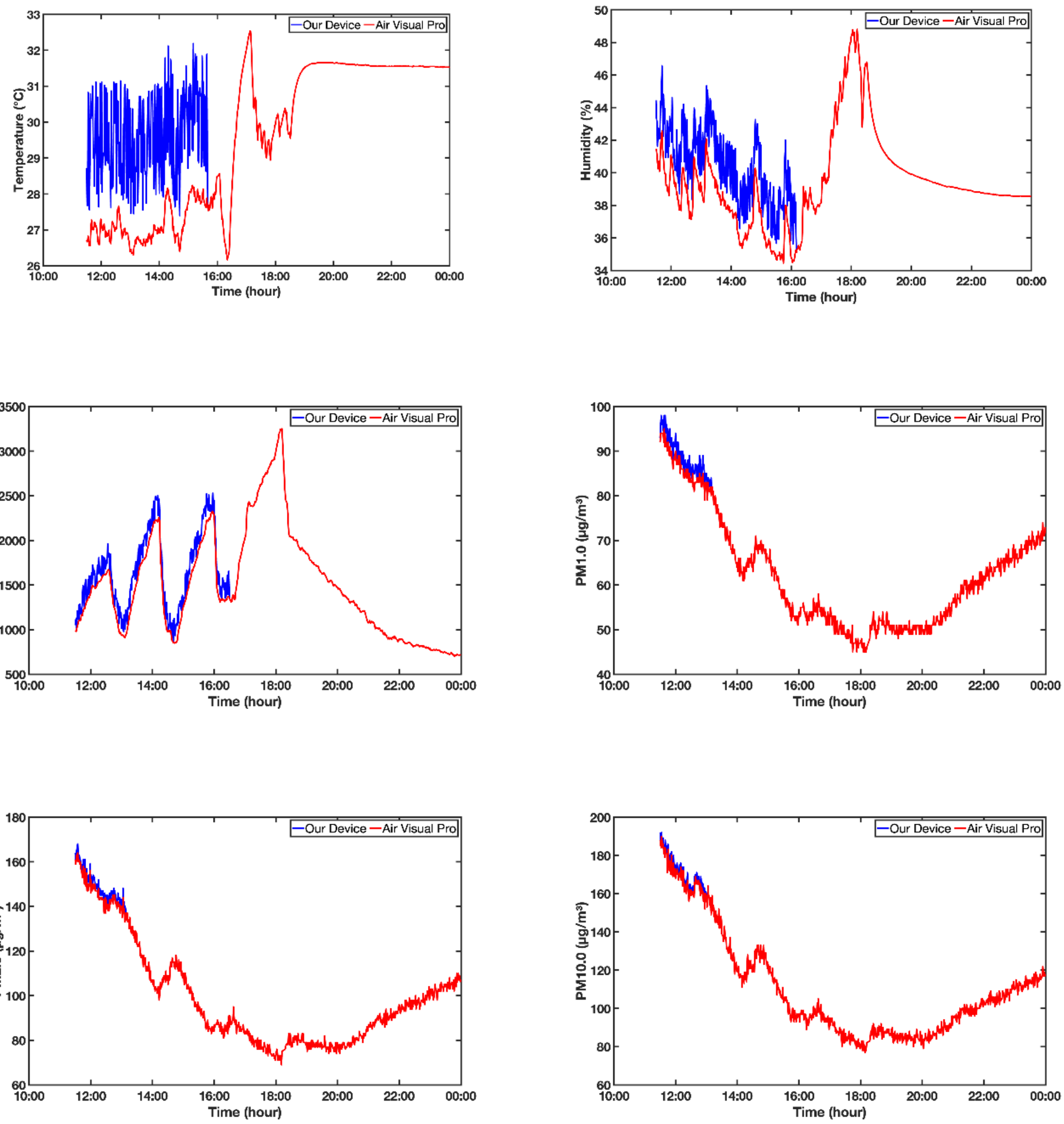


Figure 2: Calibration of our IoT sensor devices against the industry standard AirVisual Pro sensor device.

ANALYIS OF THE SENSOR DATA

We installed two IoT sensor devices in a classroom at the University in order to monitor and assess the air quality of the room. One of the sensor devices (Sensor_front) is installed on the front wall of the room and the other one (Sensor_back) on the back wall around 7 meters apart. Sensor_front and Sensor_back are situated 0.4 meter and 0.9 meter away from the side wall respectively. Sensor_front is around 0.34 meter higher than the Sensor_back. These devices are set to measure data every one-minute interval and the measured data are stored in a google sheet on the Cloud at real time through a wi-fi router.

The classroom we have selected for our study is a large room with dimensions of 12.395m by 7.0104m by 3.048m (see Figure 3). The room is facing the east direction with sun lights coming only from the north direction through a large single glazed glass window. It has only one door at the front. The classroom is airtight apart from a small ventilator with fixed blinds at the bottom of the front door. The room has two air conditioners with a cooling capacity of 10.3 kW. The air conditioners are split type and not ducted. Thus, these air conditioners cool the existing air in the room and recirculate it through filters with no option for fresh air intake from outside. The class hours are from 8:00 am to 8:00 pm during the semester and the average number of students in the classroom is 30 to 40. In the summer (usually from February to November) the air conditioners will be operated only during the class hour and will be shut off at night.

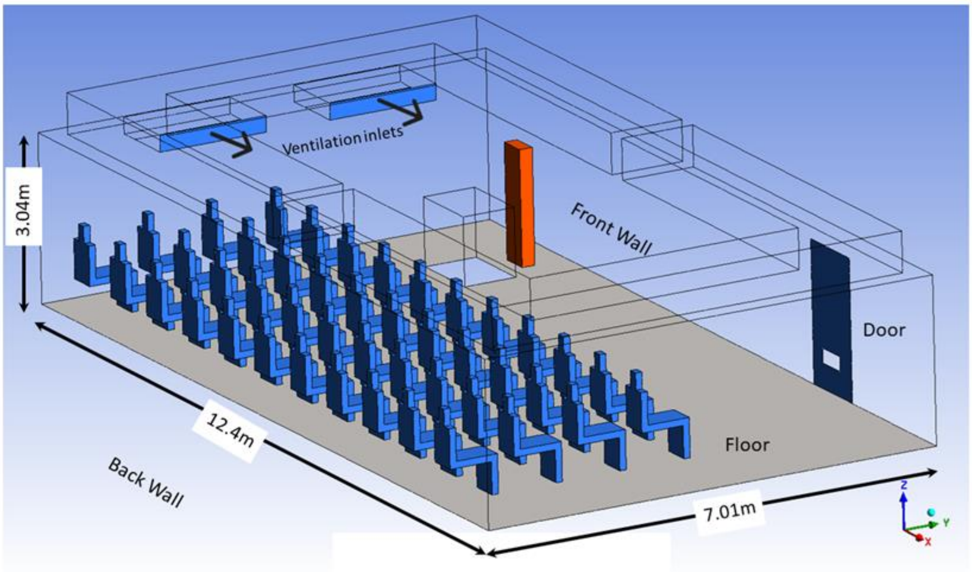


Figure 3: A 3D sketch of the classroom with the occupants (seated students and standing lecturer).

With our two sensor devices (sensor_front and sensor_back) in the classroom, we have obtained data for the following 6 parameters over a 10-month period (from March to December in 2023): temperature, humidity, CO_2, PM1.0, PM2.5, and PM10. In Figure 4 we present all the data for the above 6 parameters for the entire period of 10 months. Then parameters have been averaged out over the entire range of data for a single day and presented in Figure 5. The main trends and the patterns that can be deduced from these figures are discussed below.

From Figure 5a we observe that the temperature of the classroom remains steady at a comfortable level (less than 25ºC) during the class hour when the air conditioner (AC) remains on. This suggests that heat loss is minimum in the classroom, and it is sufficiently sealed and airtight. There is difference of around 4ºC between the values of the temperature recorded by the two sensors. Since sensor_front is stuated higher than sensor_back by 0.34 meter, it records higher temperature values. This is called thermal stratification – warm air rises up and cold air settles to the bottom. The indoor temperature is much lower than the outdoor temperature during the daytime. This is obviously due to the effects of the air conditioners. We suspect that the classroom might be cooler than necessary, and there is a scope to reduce the energy cost by operating AC at a higher temperature setting or by turning it on and off intermittently. This investigation is ongoing.

We observe an inverse relation between humidity and temperature in our data (Figure 4a and 4b) which is consistent with the available studies. The ideal humidity level is between 40% to 60%. The average humidity inside the classroom over a 24-hour period remains flat at around 50% (Figure 5b). Over the 10-month period the humidity inside the classroom fluctuates between 30% to 75% (Figure 4b).

The level of CO_2 reaches a peak level at around 1000 ppm during class hour (Figure 5c). The CO_2 level goes down to the atmospheric level (400 ppm) quite slowly when the class ends, and the room becomes empty. According to WHO guidelines the CO_2 concentration in indoor spaces should be maintained below 1000 ppm. Note that the highest peak value of 1000 ppm recorded by our sensors could be much higher if the sensors were placed at the head height of the sitting students in the classroom. The high CO_2 concentration also correlates to high infection rate. We need to employ more effective ventilation strategies in the classroom to reduce the CO_2 exposure level without compromising on the heat loss. This work is currently ongoing.

The PM particles show a sinusoidal trend inside the classroom – highest level in the morning and lowest level in the evening. This is consistent with other available studies in literature. The current WHO guidelines state that 24-hour average concentrations of PM2.5 and PM10 should exceed 15 µg/m3 and 45 µg/m3 respectively. Our data for the classroom shows average values of 65 µg/m3 and 75 µg/m3 for PM2.5 and PM10 respectively which far exceed the limit set by the WHO guidelines. This is a very serious and alarming situation. We need to come up with drastic measures to reduce the concentrations of PM particles.

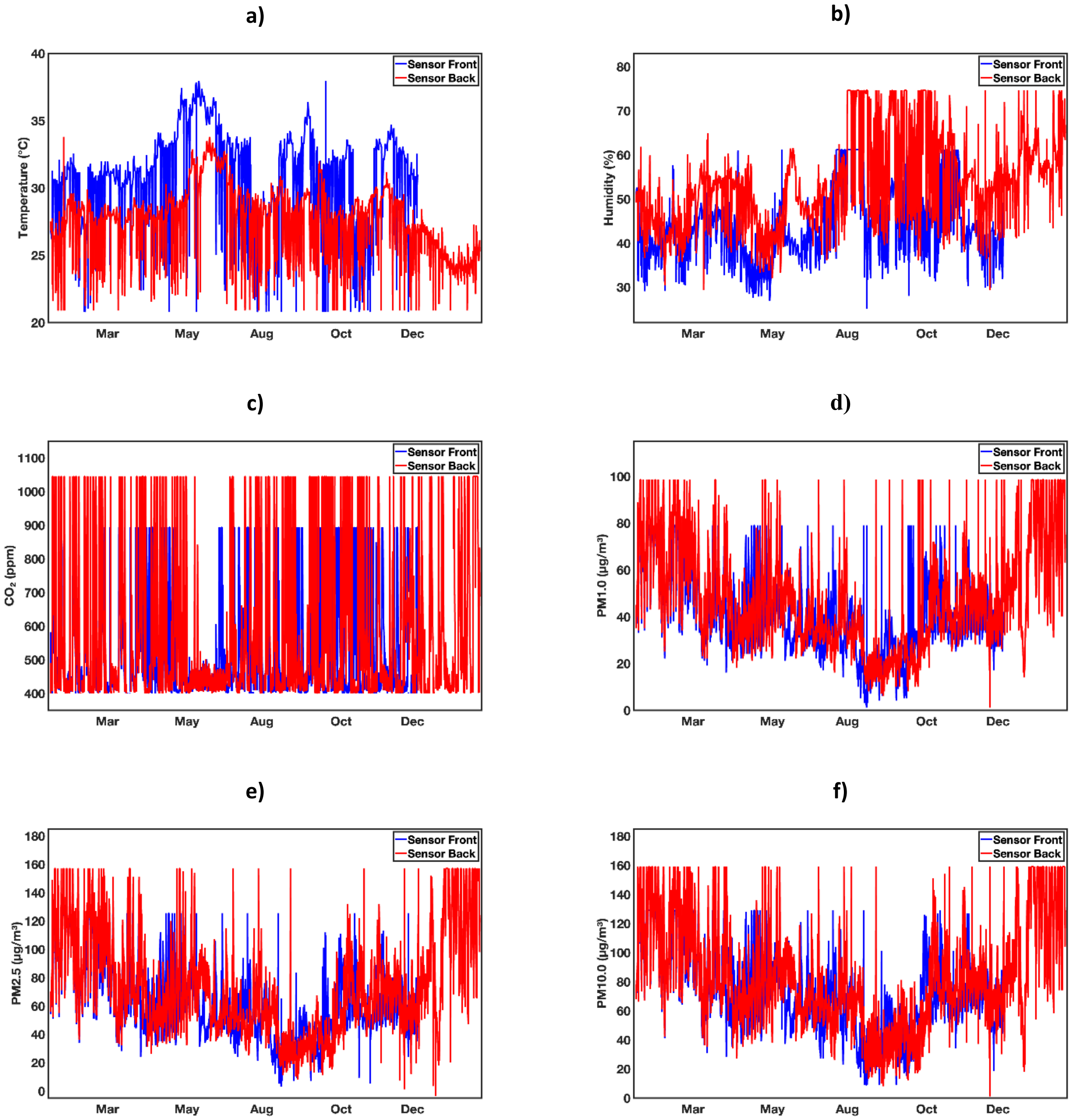


Figure 4: Data for IAQ parameters of the classroom obtained from 2 sensor devices: sensor_front (blue solid line) and sensor_back (red solid line). All the data acquired over a period of 10 month from March to December in 2023.

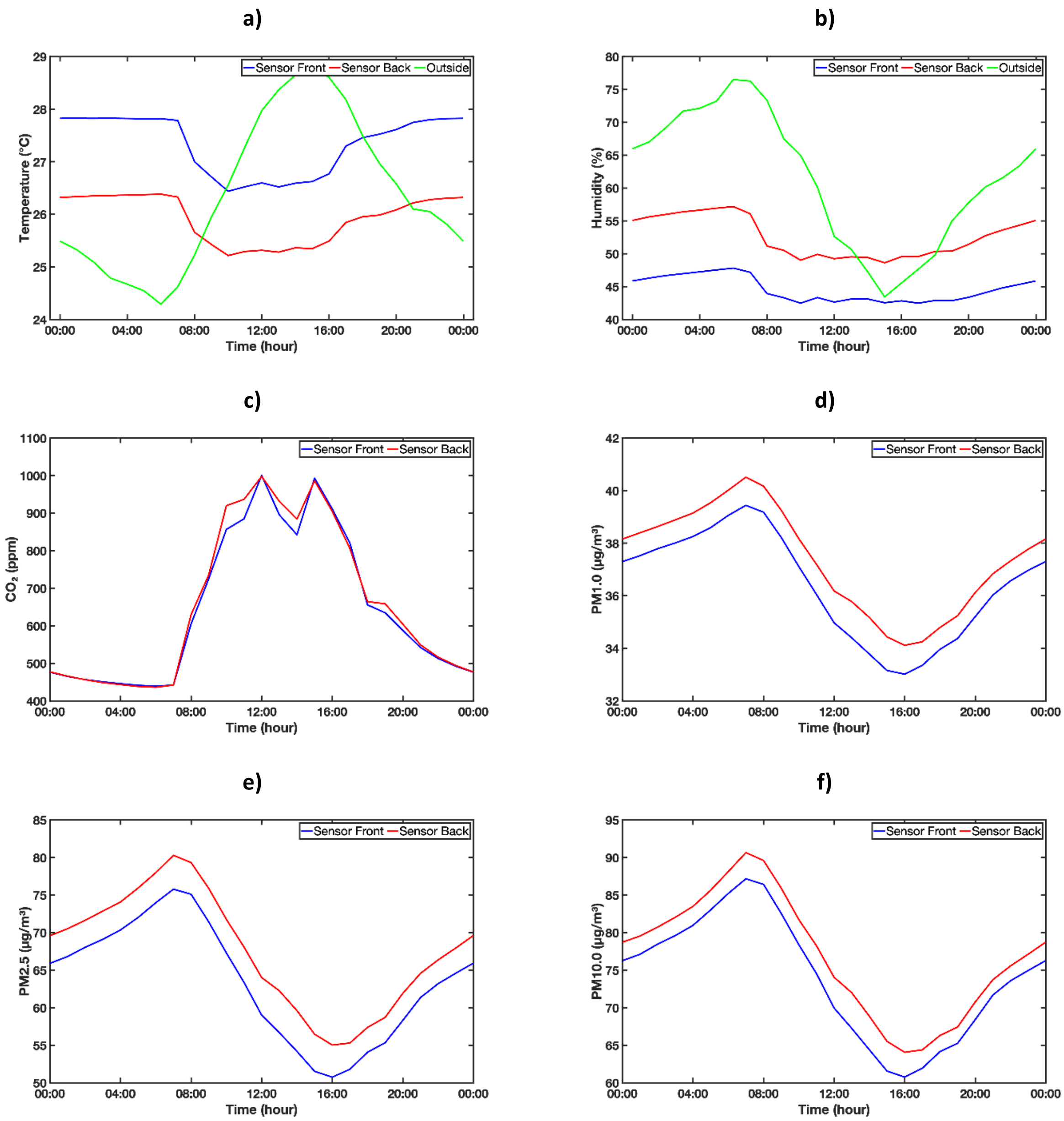


Figure 5: Average values of the IAQ parameters of the classroom for a single day for 2 sensor devices: sensor_front (blue solid line) and sensor_back (red solid line). The average was obtained from the entire range of data collected over a period of 10 month from March to December in 2023.

MODEL AND PARAMETERS FOR CFD SIMULATIONS

In this study, we use computational fluid dynamics (CFD) to study and assess the effects of various ventilation strategies on indoor air quality (IAQ), thermal comfort and infection risk in a university classroom (see Figure 3). Our chosen classroom is situated in a hot and humid environment and is representative of classroom designs widely used in higher educational institutions in Bangladesh.

A 3D CFD model was utilized to predict the air velocity, temperature and CO concentrations. We have used a finite volume-based method to solve numerically to predict the movement of air i.e. velocity, heat and pollutants (CO_2, particles) distribution in the classroom. Since the airflow in the classroom is turbulent, we use the Reynolds averaged Navier-Stokes (RANS) approach to take these flow regimes into account. In particular, we have used k-ω SST RANS model to solve the steady state turbulent airflow conditions in the classroom [1]. To include the effects of buoyancy from heat sources from the occupants and incoming cold air, we solve the energy equation with the Bousinessq assumption to couple with the momentum equation [2]. Furthermore, we solve an additional steady state transport equation to capture the dispersion of pollutants like CO_2 emitted by the occupants [3]. For steady-state simulations of the airflow, RANS equations are solved in a mesh containing approx. 5 million cells. The geometry of the classroom and the corresponding mesh generation, including the flow simulation, are conducted with the ANSYS v2022 software [4]. Appropriate boundary conditions were implemented to capture the ventilation system, heat and pollutant sources. All the geometrical and boundary conditions used in all our simulations are presented in Table 1. Students are represented as manikin simplification and surface heat flux based on the paper by Yan Y [5],

Table 1: Computational domain and simulation parameters of ANSYS CFD used to simulate the flow inside the classroom.

Parameter	CFD (SI units)
Air condition inlet flow rate	$0.778\,\frac{m^3}{sec}\ (7.8\,ACH)$
Student heat source	$22\,\frac{W}{m^2}$
Student breathing rate	$0.22 \times 10^{-3}\,\frac{m^3}{sec}\left(0.22\,\frac{Litres}{sec}\right)$
Fluid density	$1.225\,\frac{kg}{m^3}$
Kinematic viscosity	$1.46 \times 10^{-5}\,\frac{m^2}{sec}$
Thermal diffusivity	$1.963 \times 10^{-5}\,\frac{m^2}{sec}$
Room height	$3.048\,m$
Room width	$7.0104m$
Room length	$12.395m$

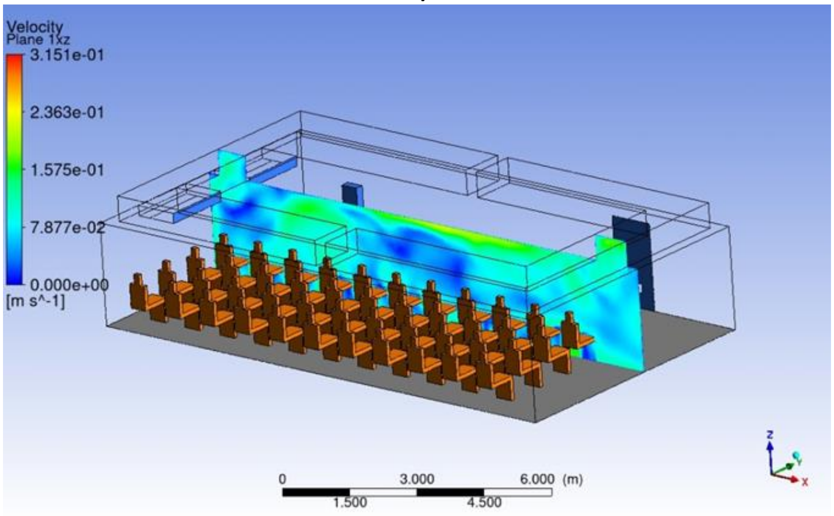
CFD SIMULATION RESULTS AND DISCUSSION

Here we present several CFD simulation results for our chosen University classroom obtained using the model and the parameters described above. The chosen parameters or boundary conditions were used to match the conditions in the classroom obtained from the IoT sensor data. Since we did not have any anemometers to measure the air velocity we have used the settings from the user manual of the air conditioner unit. The parameters (geometric, thermal and breathing rates) used for the occupants have been obtained from literature in a typical indoor setting.

Figure 6 shows the airflow speed distribution in the classroom (airflow) at various vertical slices; from the figures, we can see that the airflow in the room is not well mixed. The incoming airflow from the air condition unit is colder, heavier and faster than the ambient air and drops down Figure 7(b),(c) to the floor due to buoyancy. This effect is much clearer in Figure 7 for the temperature distribution of the room. The temperature distribution in the classroom shows the presence of stratification, i.e. the air temperature varies from a minimum value at the floor level to the maximum value at the ceiling. This behavior is expected in a room with a heat source, i.e. human occupants and the presence of incoming cool air from the air conditioner. Temperature sensor data also confirmed this behavior predicted by CFD.

Figure 8 shows the distribution of (CO_2) in the classroom. The primary source of CO_2 is the occupants in the classroom. Since the air conditioner does not introduce any fresh air into the classroom, the CO_2 concentration rises to a very high value compared to the outdoor ambient value during the class time, as shown in Figure 5 (c). These high values demonstrate poor IAQ or ventilation in the classroom. This will have a detrimental effect on the cognitive performance of the students, as evidenced in the literature. Furthermore, the risk of spread of airborne infectious disease within the classroom will be high as well. Using CO_2 concentration as a proxy for infection risk, one can calculate the risk at different positions of the room. Using CFD prediction, we will be able to explore various room ventilation strategies to find the optimal configuration for improved IAQ, thermal comfort, infection risk and energy consumption. This work is currently ongoing.

a)



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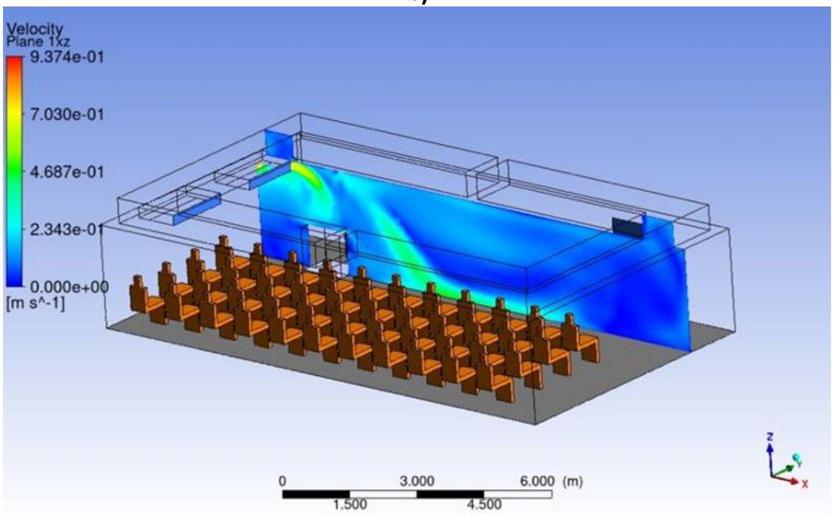

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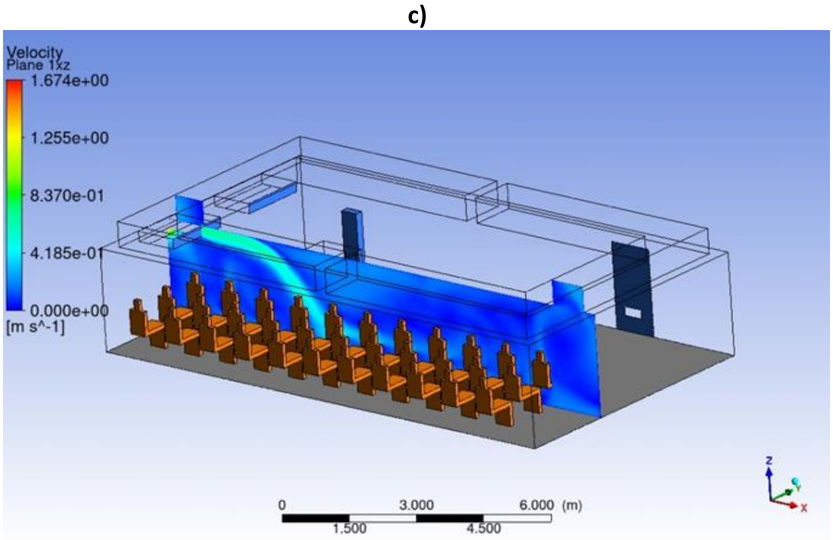
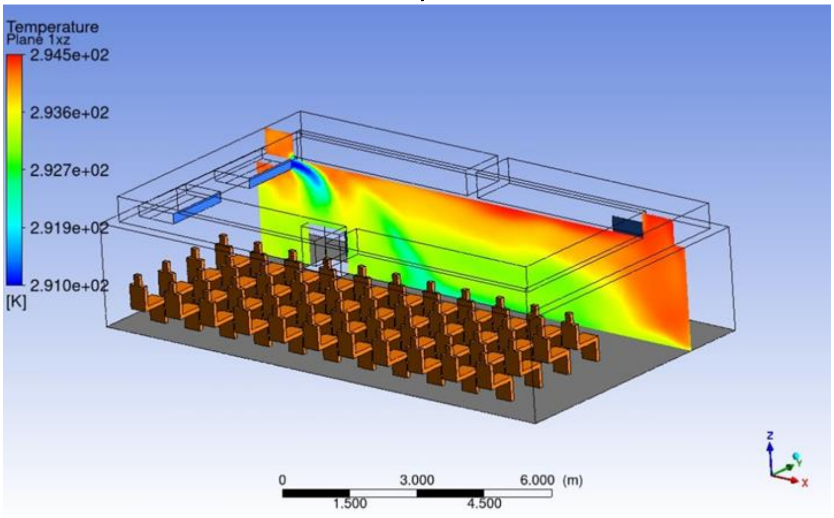


Figure 6: Contour plots of the airflow speed in the classroom at various vertical slices obtained from the steady-state CFD simulations.

a)



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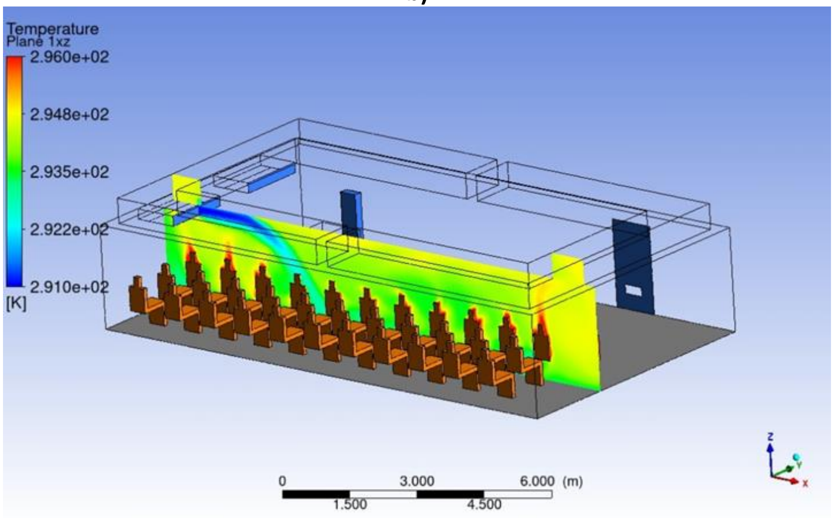
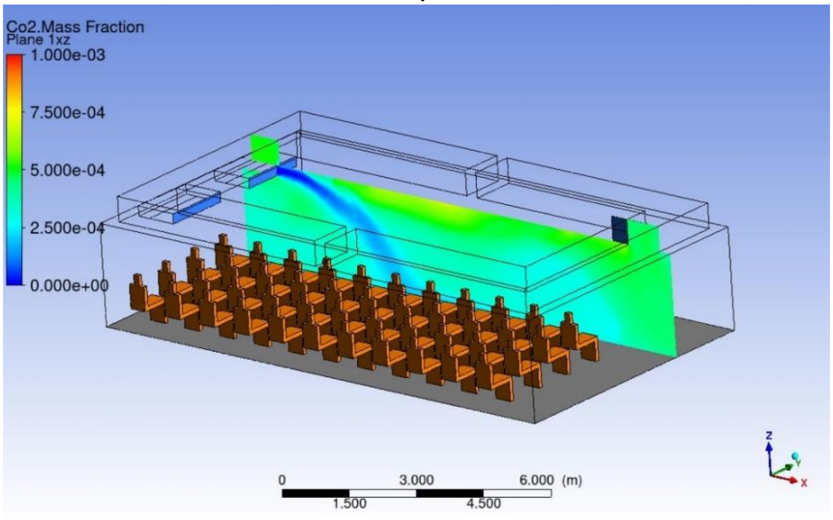


Figure 7: Temperature distribution of air in the classroom at various vertical slices obtained from the steady-state CFD simulations.

a)



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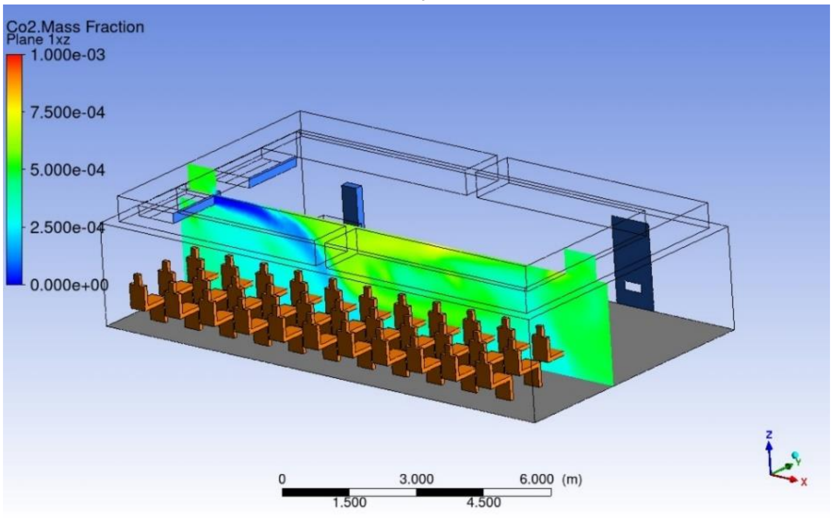

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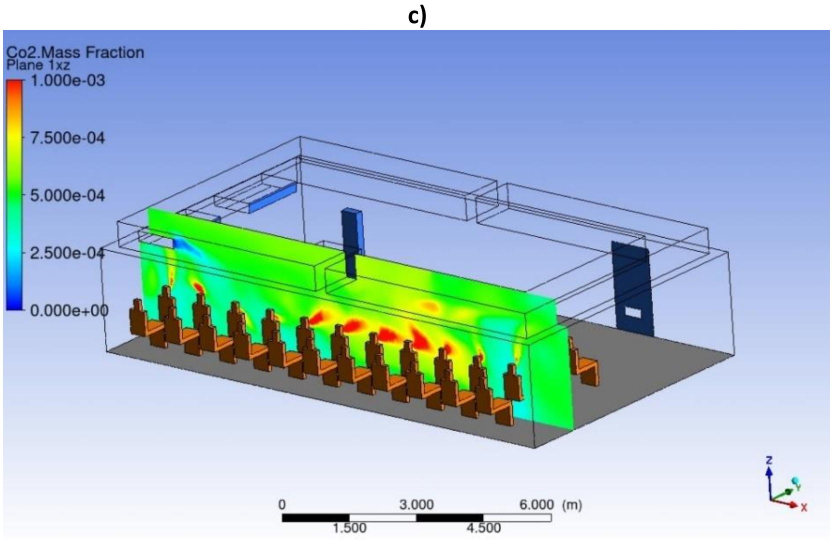

Figure 8: CO_2 mass-fraction in the classroom at various vertical slices obtained from the steady-state CFD simulations.

SHORT TERM GOALS

We plan to complete the following studies and publish a few journal articles and conference proceedings within the year 2024-25:

1) An Assessment of the indoor air quality (IAQ) and ventilation of a typical University classroom in a hot and humid climate. This study has been completed and presented in this report. We are working on an article which we plan to submit to a Journal soon.
2) A particular design optimization study: use the University classroom as a digital twin, perform extensive CFD simulations for different ventilation strategies and try to identify the most optimum condition for the indoor environment of the classroom.
3) An Investigation into the factors influencing the airborne spread of viral (SARS-CoV-aerosol particles within a University classroom. We will try to quantify the infection rate for the classroom and provide a few mitigating strategies.
4) Thermal comfort level and energy optimization. In this study we will quantify the thermal comfort level for our particular classroom and suggest a more efficient energy management strategy.
5) Predictions of indoor environmental conditions of a typical University classroom using ML (Machine Learning) models.

LONG TERM GOALS AND OBJECTIVES

The scope and the prospects of our project are very promising and far-reaching. In order to fulfil the enormous potential of this project, we plan to expand and evolve it in the following directions: firstly, we plan to increase the number of sensor devices in the classroom to enhance the spatial granularity of the measured parameters namely, temperature, humidity, CO_2, and particulate matters (PM). We are already in the process of assembling 10 new IoT sensor devices. Secondly, we want to do extensive design optimizations for retrofitting i.e. for suggesting improvements and modernizing of the existing structures (classrooms, residences, large office spaces, hospital rooms etc.) without requiring any redesigns or rebuilding retrofitting Thirdly, we intend to adopt Machine Learning (ML) models to analyze our extensive sensor data and CFD simulation results and thus help to predict indoor environmental conditions and trends. Lastly, we aim to delve into the field of Internet of Things (IoT). IoT technology can interconnect our sensors and other devices to create a unified, responsive system that can adapt to fluctuating environmental conditions.

Along with these research efforts we will certainly continue our existing collaboration with the research group at the University of Leeds, UK. We will explore other opportunities to forge internal collaborations within IUB with the ML and IoT research groups and with the Department of Environmental Science and Management. We also plan to secure external research funds to continue and expand our research efforts. We are currently exploring the prospect of writing a joint grant proposal with Dr. Khan's research group at the University of Leeds, UK.

PROGRESS TIMELINE

2021

Initiating discussion with a research group led by Dr. Khan at University of Leeds, UK about a suitable research project on the IAQ of Bangladesh; Assess the importance and viability of such research in the context of Bangladesh; Submitted an application for Sabbatical Leave to the IUB authority.

2022

Purchased and built a high configured desktop in March to run the time-consuming and very expensive CFD simulations required for this project; Sabbatical application approved, leave granted for two terms (8 months) from May 2022 to December 2022; Background preparation (literature review, basics of CFD, acquiring sensor component, building two IoT sensor devices, calibrating) for the project started from May; Regular weekly online meeting with the external collaborator in UK.

2023

Two IoT sensor devices installed in the classroom BC6007 on February 25 and real-time data acquisition started from that day; ANSYS software package for CFD simulation installed, getting familiar with the ANSYS package running toy calculations; Generating CFD simulation results for classroom BC6007; Analyzing sensor data; Complete an assessment of the IAQ and ventilation of classroom BC6007; Regular weekly online meeting with the external collaborator in UK and the researchers of IUB working in this project.

2024-25

Try to achieve the short-term goals mentioned above: Crucial period for our project, Main target is to produce tangible outcomes in terms of journal publications and conference presentations.

2025 and beyond

Try to achieve the long-term goals and objectives mentioned above; Main target is to acquire external research fund.

CONCLUSION

We have initiated a collaborative project to study the indoor environment in Bangladesh. As a first step of our study, we have concluded an assessment of the IAQ of a generic University classroom in Bangladesh using both real-time sensor data and steady-state CFD simulations. Our study shows that the classroom is sealed tight quite effectively so that the loss of heat is minimum. However, this creates an issue with a high build-up of CO_2 in the classroom during class hours when the occupancy is high. This in turn increases the infection rate and enhances the risk of infectious diseases. We need to adopt a more optimum ventilation system and more efficient energy management in the classroom. That is the topic of our next study. We are expanding our studies by deploying more sensor devices in the classroom and by adopting predictive machine learning models. The collaboration with the research group at the University of Leeds has been immensely beneficial and fruitful. We plan to keep the collaboration going for the foreseeable future.

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